Automated Brand Safety Classification with Vision Transformers

A multi-class visual moderation pipeline for programmatic display advertising.

1. Executive Summary

Digital publishers make money through programmatic advertising. However, automated ad delivery allows unsafe ads to slip past basic filters, which harms platform integrity. This project creates an automated machine-learning solution for multi-class brand safety classification using Vision Transformers (ViTs).

The system scans display ads to identify restricted content, including firearms, explosives, and unregulated financial assets like cryptocurrency scams. ViTs use self-attention to balance global context with precise object detection. Developed and tested locally on an Apple M2 Mac Pro with 32 GB of unified RAM, it produces a scalable, low-latency prototype that safeguards publisher reputation and user experience.

2. Problem Statement

Programmatic platforms serve billions of visual ads daily without direct publisher oversight of each ad. Non-compliant advertisers take advantage of this to distribute NSFW imagery, weapons, explosives, or misleading financial schemes.

Traditional defenses depend on metadata keyword blocking or basic binary vision models. They overlook visual violations that don't have textual labels, and standard convolutional networks fail to detect obscured objects since they lack global context. A brand safety breach can drive premium advertisers away, damage reputation, and create regulatory risks. Publishers need a smart, multi-class visual moderation pipeline that effectively separates safe content from various types of non-compliant imagery.

3. Proposed Solution (ML Pipeline)

The solution is an end-to-end inference pipeline anchored by a fine-tuned Vision Transformer. ViTs handle images as patches and use self-attention to map long-range dependencies across the visual field, making them more effective than localized CNNs in cluttered ad layouts. The execution uses PyTorch with Metal Performance Shaders (MPS) on the M2 Mac Pro; the 32 GB of unified RAM comfortably manages large weights and batch sizes without slowing down.

Ingestion & pre-processing: assets are resized to 224×224, normalized, split into 16×16 patches, and flattened to linear embeddings with positional encodings.

Model architecture: a pre-trained ViT-B/16 backbone passes embeddings through multiple Transformer encoder layers to capture semantic context.
Classification head: a custom linear layer with Softmax outputs a probability distribution over four classes: Safe, Firearms, Explosives, Unregulated Financials.
Policy engine: a deterministic module flags, blocks, and replaces any ad if its restricted-class probability exceeds a certain threshold.

4. Objectives & Scope

Primary objectives:
Multi-class model with a macro F1-score above 0.93.
False positives under 2% to protect legitimate ad revenue.
Fast inference for real-time web delivery.

In scope:
Static display ads (JPEG, PNG) trained and validated locally on the M2 Mac Pro.
An operational prototype available via a local API with visual attention maps.

Out of scope (this phase):
Video or animated-GIF ads, NLP/landing-page analysis, and direct integration into live RTB exchanges.

5. Data Strategy & Datasets

Training combines open-source repositories, targeted scraping, and synthetic augmentation across the four classes. Public datasets below provide examples for each category; a scraped set of promotional crypto tokens, speculative trading charts, and high-risk financial ad templates adds to the financials class, while benign display ads collected from news, e-commerce, and lifestyle sites represent everyday traffic.

Class	Dataset	Source / URL
Firearms & Weapons	OD-WeaponDetection (UGR, pistols & knives, classification + detection)	github.com/ari-dasci/OD-WeaponDetection
	UGR Weapons Detection Open Data	dasci.es/transferencia/open-data/24705
	Kaggle Weapon Detection (9 weapon types, YOLO labels)	kaggle.com/datasets/snehilsanyal/weapon-detection-test

	Roboflow Weapon Detection (9.7k images, 28 classes)	universe.roboflow.com/.../weapon-detection-m7qso
Explosives	Kaggle Explosive Detection	kaggle.com/datasets/saurabhshahane/explosive-detection
	Open Images V7 (grenade, weapon, tool categories)	storage.googleapis.com/openimages/web
Unregulated Financials	Roboflow Cryptocurrency image dataset	universe.roboflow.com/helloo/cryptocurrency-wga6l
	Mendeley Cryptocurrency Scam Dataset	data.mendeley.com/datasets/ffd5crf8mx
Safe / Benign & NSFW reference	COCO (everyday objects for benign backgrounds)	cocodataset.org
	deepghs NSFW detection dataset (Hugging Face)	huggingface.co/datasets/deepghs/nsfw_detect
	Falconsai NSFW ViT model & data reference	huggingface.co/Falconsai/nsfw_image_detection

Since unsafe ads make up a small part of the traffic, augmentation (such as rotations, color jitter, and cropping) increases the size of restricted classes. Focal Loss prevents the Safe class from dominating gradient updates.

6. Market & Stakeholders

In AdTech, brand safety is now a basic requirement. Supply- and demand-side platforms and premium publishers make up the core market. Key stakeholders include:
Publishers: protect brand value and audience trust without heavy manual review.
Advertisers: premium brands seeking clean environments for their ads.
AdOps teams: transition from manual review queues to exception-based auditing.

7. Value Proposition & Differentiation

The main value lies in shifting from keyword/binary filtering to context-aware multi-class categorization. ViTs can detect dangerous objects even when they are stylized, partially obscured, or embedded in busy layouts. Unlike CNNs that only analyze localized pixel clusters, ViTs maintain a holistic view, and their attention maps provide clarity: teams can see which patches triggered a flag, simplifying debugging and reducing the risks associated with black-box systems.

8. Business / Operating Model

The system operates as intelligent middleware between the ad server and the client browser, acting as an internal microservice or a containerized module deployed on GPU-optimized infrastructure. It scales under a high-volume, low-margin model and can charge based on processing tier or API volume. This phase simulates the model locally on the M2 Mac Pro to demonstrate feasibility before deploying to enterprise cloud systems.

9. Social Impact

Automating brand safety reduces the financial appeal of malicious actors and protects users from unverified sales of weapons, explosives, and crypto fraud. It also spares human moderators from the psychological burden of constant exposure to graphic content, limiting human intervention to uncertain edge cases.

10. Success Metrics

Model performance:
Precision > 98% (Safe class): ensure compliant ads are never blocked, protecting revenue.
Recall > 95% (restricted classes): capture nearly all non-compliant assets.
Macro F1 > 0.93: ensure balanced performance across all categories.

System performance:
Latency below 50 ms per asset; strong throughput under concurrent load.
Resource usage is minimal on MPS, showing efficient use of the 32 GB unified memory.

11. Deliverables

A documented Python repository (training, pre-processing, validation) optimized for Apple Silicon MPS.
A curated, balanced, labeled validation/test dataset registry.
Trained model weights in optimized serialization formats.
An interactive local dashboard for uploads, probability outputs, and attention maps.
A technical capstone report detailing methodology, experiments, tuning logs, and results.

12. Risks & Mitigation

Data scarcity for critical classes. Explosives and specific scams are rare. Mitigation: use aggressive augmentation, synthetic generation, and class-weighted loss.
Transformer latency. ViTs require significant computation. Mitigation: employ post-training INT8 quantization for faster inference with minimal impact on accuracy.
Adversarial evasion. Attackers may add noise, borders, or distortions. Mitigation: implement robust training transformations (Gaussian blur, contrast shifts, random crops) to ensure the model focuses on the essential objects.